Data Mining Project

MASTER DEGREE PROGRAM IN DATA SCIENCE AND ADVANCED ANALYTICS

Customer Segmentation: Insurance Company

Group R

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January, 2023

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1. Introduction

The goal of this project is to cluster the customers of A2Z Insurance in order to better understand their needs, preferences, and behaviors. By leveraging clustering techniques, we can group the company's customers into distinct clusters that share similar characteristics, such as income levels, buying habits, and other relevant data. This will allow the company to better target their clients with the right products and services, thus maximizing their profits and resource usage efficiency.

To achieve this goal, we will need to leverage the company's data, which will include socio-demographic and behavioral information, including data regarding their insurance policy spendings. With this information, we can develop an analysis that will identify each customer's cluster and allow the company to adapt their marketing strategies accordingly.

With the use of historical data on the clients, we hope to create clusters that reflect the overall insurance-buying population of A2Z Insurance.

2. Data Exploration

Our first step was to perform an initial exploration of the available information and try to identify data patterns, structural problems and draw early conclusions. We started by uploading our data file into a pandas data frame and creating a copy of our original data information. The features we had were 'CustID' - an ID that specifically identifies a costumer, 'FirstPolYear' - the year of the customer's first policy, 'BirthYear' - the year when the customer was born, 'EducDeg' - the academic degree of the customer, 'MonthSal' - gross mothly salary, 'GeoLivArea' - the area of residence, 'Children' - a binary variable that represents if the customer has children or not, 'CustMonVal' - the lifetime value of the customer, 'ClaimsRate' - a metric that results from the amount paid by the insurance company divided by the premiums that the customer pays, and 'PremMotor', 'PremHousehold', 'PremHealth', 'PremLife', 'PremWork' - all the premiums that the customer may pay. In total, we had 10296 observations.

We started by transforming the 'CustID' variable into an int64 and then we defined that variable as our dataframe index (considering it is unique and it does not have null values).

Afterwards, we used descriptive statistics and data visualization tools (such as pandas profiling) to generate an overall report of the data. The description of the data frame - figure T2.1 - showed us the presence of possible outliers in some of the features where the where there was a big difference between the mean and the media, and extreme minimum/maximum values. Some inconsistencies and potential insights were also found, such as:

- 3 duplicate rows;
- 'BirthYear' and 'MonthSal' have a strong negative correlation (using Spearman) showing that older people usually have a higher salary;
- 'GeoLivArea' is not correlated with any feature (using Kendall and Cramér) and we do not have any information about the meaning of the area codes. Thus, it does not seem to be a good variable to use in the clustering part;
- 'CustMonVal' has an even stronger negative correlation (using Spearman) with 'ClaimsRate';
- All the features corresponding to premiums are correlated with each other (again using Spearman);
- Almost all features have missing values and 'PremLife' had the most (104), representing 1% of this feature data;
- 'CustMonVal' is highly skewed to the left. 'FirstPolYear', 'ClaimsRate', 'PremMotor', 'PremHousehold' and 'PremHealth' are highly skewed to the right. This called our attention to the possibility of having outliers in these features;
- The mean and median for 'PremHousehold', 'PremLife' and 'PremWork' are very different, indicating also the presence of outliers;
- 'BirthYear' had an extremely low value of 1028 that needed to be treated;
- 'FirstPolYear' was smaller than 'BirthYear' in 1997 ocasions. *Per si*, this meant that around 19% of our population already had an insurance policy before they were born;

- There were occurrences of people born in 2001 (15 years old) with unusualy high ‘MonthSal’ for that age;
- The percentage of customers with a PhD was very low (around 7%).

These potential problems will be solved in the data preprocessing stage. We then confirmed the datatypes were appropriate for our purposes and created a table with the most common descriptive statistics for all variables to systematize the variables’ information. Finally, we divided the features into ‘non_metric_features’ and ‘metric_features’.

3. Data Preprocessing

Regarding the preprocessing stage, we corrected some of the incoherences/inconsistencies identified above (or decided that they were not incoherences but instead a normal part of the data), removed duplicate rows, encoded categorical data, renamed some columns, imputed missing values, and removed outliers. To ensure flexibility of our preprocessing strategy throughout the project, we also made sure to create numerous copies of the different versions of the dataset.

3.1. Inconsistencies/Incoherences and Duplicates

We started by dropping all the duplicate rows and replacing the 'FirstPolYear' values that did not make sense into missing values (in this case, there was only one – 53784). Then, we confirmed that we had 1997 rows where the 'FirstPolYear' was below 'BirthYear', which was a strong inconsistency. We also confirmed that there were unreasonable values for the salary for the people born in 2001. Instead of transforming all this data, as it was too much, we decided that it would be better to drop the variable 'BirthYear'. We assumed that the information provided by the company is probably more accurate than the information provided by the client and we also considered the fact that the A2Z Insurance databases are poorly maintained.

3.2. Data Imputation

Our first step was replacing all the missing values in premium variables with the value 0. We assumed that the premium variables without any value meant that the client had not acquired the product (and, in fact, there were not any 0 values in the premium variables' values).

For the other missing values, we then experimented with the following four different imputation methods.

3.2.1. Median and Mode

Using the mode, all missing values in 'Education' were replaced by "b´3 – Bsc/Msc", in 'GeoLivArea' by 4.0 and 'Children' with 1. In the numeric variables with missing values – which excludes all premium variables - all missing values are replaced with the 50th quantile value of the respective variable.

3.2.2. KNN Imputation with weights="uniform" and Mode

Since it is a distance-based method and each variable has its own unique scale we must normalize the data, which we did by applying the MinMaxScaler. The KNN can only be used with numeric features, and we applied it using the Euclidean distance and 5 neighbors as parameters. After inputting all metric missing values, we reverted the scale to retrieve the original values. We then used the mode to impute the missing values in the non-metric features.

3.2.3. KNN Imputation with weights="distance" and Mode

The difference in the weights makes it an interesting option since it helps to reduce the influence of outliers and focuses the prediction on the more "representative" data points.

3.2.4. Iterative Imputer using Linear and Logistic Regressions as Estimators

As we had missing values in both metric and non-metric features, we firstly imputed the ones for the metric features using Linear Regression and secondly the ones for the non-metric features using Logistic Regression. For the first, we fitted and transformed the iterative imputer using only the metric features and for the second we fitted and transformed using only the non-metric features.

After some trials, we decided to use KNN inputer with weights=distance and mode as our final choice as it provided good results in the clustering and is generally a robust option to impute missing values. We were also not sure if the non-metric variables were capable of being good estimators for the missing values, thus us not using the iterative imputer (despite knowing that its initial strategy is the mode).

3.3. Outlier Removal

When preparing data, detecting, and processing outliers is an important step to consider. To ensure the data is treated in the most beneficial manner, a particular treatment must be adopted. We tried six different ways to detect outliers and then chose the most effective one and with as little loss of relevant information as possible.

We started by encoding the 'EducDeg' variable because we would need it to use the isolation forest method and it would also be useful for the rest of the project.

3.3.1. Manual

We defined filters based on visualizations of our data. We kept 99.18% of the original data.

3.3.2. IQR

In this method we defined an upper and lower bound using the inter quartile range (IQR) multiplied by 4. We chose the value 4 based on visualizations of our data and the percentage of data that was kept. We then excluded all observations outside of that boundary and kept 98.26% of the original data.

3.3.3. Manual and IQR

Here we combined the two previous methods and kept 99.92% of the data.

3.3.4. Z-Score

Z-score standardizes the data and removes all values that are 3.5 standard deviations apart from the mean (0). We chose 3.5 for a similar reason to the value in the IQR method. We kept 97.49% of the data.

3.3.5. DBSCAN

Using DBSCAN we excluded all points that are not considered core or border. Firstly, we had to determine the optimum epsilon to use as a parameter (0.1) using the Elbow Method and considering both visualizations of our data and the percentage of data that was kept - figure F3.1 . We kept 98.83% of the data.

3.3.6. Isolation Forest

We defined the contamination as 0.03 as we intended to exclude 3% of the data. We chose 3% considering, again, visualizations of our data and trying to transform as little as possible of our dataset. We kept 97% of the original data.

We decided to use Z-Score as our final choice as it seemed to exclude all the most severe outliers and provided good results in the clustering section.

3.4. Analysis of the Dataset after Outlier Detection and Data Imputation

We now created two new pandas profiling reports in order to visualize the data set after this pre-processing. We concluded that all major outliers had been removed.

3.5. Feature Creation

After completing data exploration and preprocessing, we performed feature engineering.

Here, we created the variable 'YearsCustomer', which tells us the current age of the person by subtracting the 'FirstPolYear' from 2016, as this was the current year of the database. This feature will then tell us for how many years has a client been a customer of the insurance company.

We then created the variable 'CustValYear' by dividing the 'CustMonVal' of a customer with 'YearsCustomer'. This indicates to us how much a client is worth to the company, but taking into consideration how many years the company has had him as a client (it corresponds to the value per year of being a customer).

Finally, we created the 'AnualSal' variable from the 'MonthSal' of each customer, simply by multiplying it by 12.

We had also created variables regarding the ratio of premiums paid against the total premiums paid by the client and their annual salary but we ended dropping those variables on the feature selection stage. With that in mind, we commented out their creation.

3.6. Feature Selection

We started by dropping 'FirstPolYear' and keeping 'YearsCustomer' in order to avoid redundancy. As well as that, we dropped 'MonthSal' and kept 'AnnualSal' to maintain consistency, considering that premium values are also yearly.

We also dropped the created ratio variables, as mentioned above, because we preferred using the original values (the ratio would cause the variables to be more correlated because if one has a percentage of 40%, the sum of the others would have at maximum 60%, which does not happen with the original variables). Having 'AnnualSal' and the premium of each category also gives us more information than using 'TotalPremiums'.

3.6.1. Variance of the Metric Features

We started by scaling the data using MinMax scaler in order to compare variances. There were no univariate variables.

3.6.2. Association Measures

We plotted the correlation matrix (using spearman) - figure F3.2 - and generated a profile report (using pandas profiling) with the pearson, spearman, kendall, phi and cramers correlations. 'CustMonVal', 'CustValYear' and 'ClaimsRate' have extremely strong correlations, indicating that we have a problem of redundancy (using Spearman). 'YearsCustomer' In addition, 'GeoLivArea' has both the problem of no correlation with the other variables and the lack of meaning of the area codes. Therefore, we decided to test in the clustering section if we should use or not these variables.

3.7. Feature Scaling

In order to test different feature scaling techniques during the modelling stage, we scaled the dataset using three methods: MinMax Scaler, Standard Scaler and Robust Scaler.

We choose the standard scaler as our final selection because of the results in the clustering section.

3.8. Detection of the Remaining Outliers in our Dataset (using DBSCAN)

We started by creating a new profile report (using pandas profiling). We then ran a DBSCAN with an epsilon parameter of 2 (chosen taking into account the percentage of data kept and the Elbow Method) and we kept 95.96% of the data in our final dataframe (figure F3.3.1). This was done because after starting the clustering we noticed there were still outliers (in part because of the created variables). Then, we plotted the histograms of all variables and the box plots of all metric variables (figures F3.3.2 and F3.3.3). We also plotted the pairwise relationships of the numeric variables and of all the variables. (figure F3.3.4 and F3.3.5) The dataset was ready for the clustering.

4. Clustering by Perspectives

At this stage, we split our variables into two distinct perspectives: Demographic and Behavioral. We then applied different algorithms to each and finally merged the perspectives to obtain our final solution.

4.1. Demographic perspective

The demographic perspective of our clients contained initially the variables 'AnnualSal', 'HighSchool', 'BSc/MSc', 'PhD', 'Children' and 'GeoLivArea'. We used one model to cluster our records and after some trial and error we dropped the variable ´GeoLivArea´ because it was not returning relevant information to our clusters (as expected).

4.1.1. K-prototypes

The K-prototypes algorithm was implemented in order to identify clusters from the Demographic Perspective data set, which contains both categorical and numeric values. This algorithm is an improvement on both k-means and k-modes, as it can handle both types of data. Two distinct methods could be selected as parameters to initialize the clusters: Huang and Cao. The Huang method selects the first k distinct records from the data and then assigns the most frequent categories equally to the initial k-modes. Similar results were obtained when using the Cao method which is more complex, selecting the prototypes for each object based on the density of the data point and a dissimilarity value (nota de rodapé). We decided to use Huang.

In order to identify the ideal number of clusters, we ran the algorithm from 1 to 6 clusters and plotted the clustering cost for each. Upon analysis of the plot (figure F4.1) and using the Elbow Method it appears that 2 or 3 clusters is the optimal number of groups to form and upon analysis of the created clusters we decided to use 3 as our final choice.

It is important to note that our categorical features only assume values of 0 or 1, so our centroids will always be some value ranging from 0 to 1 representing the proportion of individuals with that characteristic (for instance, children) in that cluster.

4.1.2. Cluster Profiling

By analyzing the centroid of each cluster (figure T4.1 and F4.2), it is possible to define three distinct groups. Cluster 0 is the biggest in terms of individuals and is formed by clients who have an average 'AnnualSal', mostly an 'EducDeg' of 'BSc/MSc' and most have children. Cluster 1 is composed by people with below average salary, mainly just High School education and with children. Finally, cluster 2 is formed by people with above average salary and levels of education averaging high school or BSc/MSc, but most do not have children.

4.2. Behavioral Perspective

For the behavioral perspective of the data, we started with the variables 'CustValYear', 'PremMotor', 'PremHousehold', 'PremHealth', 'PremLife', 'PremWork', 'CustMonVal', 'ClaimsRate' and 'YearsCustomer'. 'ClaimsRate' and 'YearsCustomer' ended up being dropped as they were not providing differentiating information of our clusters. We also dropped 'CustMonVal' as 'CustValYear'

was better in terms of the interpretation and was also yearly. These tests were done using K-means, which we describe below.

4.2.1. K-means

To run our k-means algorithm, we started by calculating the inertia and silhouette scores of our solutions varying between 2 and 15 clusters. We then analysed the plots and concluded that 3 clusters were the optimum level for this method, as it was the elbow of the inertia plot, 4 clusters would reduce the silhouette score and the cluster profiling had insightful information.

We selected the features that would increase the separability and the information of our cluster profiles and used them for the models that followed (considered the cluster profiles and the variations in the R^2). We kept the features 'CustValYear', 'PremMotor', 'PremHousehold', 'PremHealth', 'PremLife' and 'PremWork' and we obtained the clusters in figure T4.2. We also obtained an R^2 of 0,4729 on this solution.

4.2.2. Hierarchical clustering and K-means

To use this model, we started by computing the K-means algorithm with 20 clusters. Then, we plotted the R^2 for each linkage method and decided to use the ´ward' method. After analysing the dendrogram and interpreting the cluster profiles, we concluded that 3 clusters was the best solution.

We obtained the following cluster profiles (figure F4.7) with an R^2 value of 0,4312.

4.2.3. Gaussian Mixture model

For this method, both the BIC and AIC plot suggested that 4 clusters was the best solution. However we decided to keep 3 as it was still a good solution and would allow us to compare with the other models.

We obtained the following profiles (figure F4.9) with an R^2 value of 0,4109.

4.2.4. Self-organizing map and K-means

To compute the self-organizing maps algorithm, we started with 400 initial neurons. We used the parameter 'rect' instead of 'hexa' because although hexagonal lattices are usually able to handle more complex datasets than rectangular lattices, we were obtaining a smaller quantization error.

We then ran the k-means algorithm and plotted both the inertia and silhouette score of each solution, concluding that three clusters were still the best solution.

We obtained the following cluster profiles **(**figure F4.12) with a R^2 value of 0,4510.

4.2.5. Self-organizing map and Hierarchical Clustering

To compute the self-organizing maps algorithm, we started we 400 initial neurons. We used the parameter 'rect' instead of 'hexa' for the same reason presented above.

After analysing the dendrogram, we concluded that 2 or 3 clusters would be the optimum solution and once again we calculated for 3 in order to be able to compare model solutions (and considering that we had as before good cluster profiles).

We obtained the following profiles (figura) with and an R^2 value of 0,4184.

4.3. Cluster Profiling

Our final choice for the behavioral perspective was the self-organizing maps and k-means solution. All models had similar cluster profiles with the 3 clusters. However, self-organizing maps and k-means offered one of the best R^2 values and the profiles were more separated and clearer to interpret.

Our cluster 0 contains the companies' clients that have a below average 'CustValYear', which represents their worth to the company. These clients mostly acquire health premium policies above the other groups and car insurance policies below the average. They then spend around the average on the variables 'PremHousehold', 'PremLife' and ´PremWork´. This cluster contains slightly less customers than cluster 2 but it is quite near it as the cluster with the most indivivuals. Cluster 1 contains the companies' higher value clients that on average spend a lot more on their house, life and work insurance. They spend quite a lot below average on their motor premiums and spend around the average amount on their health policies. This is the cluster with the least number of individuals, which makes sense as it contains the most valuable customers. Cluster 2 is the group with the most people in it as contains the average value customers. This group spends the most on their motor insurance and below average on all other insurance policies.

4.4. Merging Perspectives

On the final stage of our analysis, we were left with a total of 9 possible clusters, 3 from the demographic perspective and 3 for each of the latter from the behavioral perspective. To obtain a better solution, we needed to merge the perspectives.

4.4.1. K-prototypes

We decided to use the K-prototypes model because similarly to the demographic perspective we had both numerical and categorical variables. It is important to note that, once more, we computed the cost of each cluster and, by analyzing the results and the profile of each cluster, concluded that the ideal value should be six clusters. Our data points for the model were our centroids from the previously obtained solutions. As a result, we got the following cluster profiles on figure F4.18. As before, our categorical features only assume values of either 0 or 1.

4.5. Cluster Profiling

We obtained six final clusters (decided taking into account again the Elbow Method, the cluster profiles and the context of our problem – differentiating customers to develop more focused programs).

Cluster 0 contains 2488 clients with slightly below average 'CustValYear'. These clients spend quite below average on their motor insurance premiums and around average on their house, life and work policies. They spend higher than average on their health insurance. These clients have a quite below average annual salary and their education is averaging on HighSchool or BSc/Msc and most have children.

Cluster 1 contains 1739 clients with also slightly below average 'CustValYear'. This customer has similar characteristics to the cluster 0 when it comes to policy acquisition behavior and education level. However, they have a very high average salary which differentiates them strongly.

When analyzing the variables that had been dropped, we can see that both these groups have the highest claims rate, which is reasonable considering the fact that they mostly acquire health insurance (which is possibly the most claimed insurance on the market). This makes sense as to why they have slightly below average value for the company as clients.

To target both these groups, we must have slightly different strategies. Cluster 0 clients have a quite below average salary which leads us to believe that they are not in a position to acquire a lot more insurance policies. However, it may be a good idea to make a campaign that targets them into buying motor insurance, as it usually the next insurance in line when it comes to priorities. This campaign may be a mass one such as tv or radio commercial as both cluster 0 and 1 may be targeted. Clients from cluster 1 may be targeted more specifically with publicity for buying life and house insurance as they have more financial possibilities to acquire diverse insurance policies.

Cluster 2 contains 931 individuals that have average customer value for the company. Their insurance portfolio contains above average motor premium payments and quite below average on all other insurance categories. This cluster represents people with a superior education and it is worth noting that it is the cluster with highest proportion of individuals with a PhD. Their annual salary is quite above average and the proportion of individuals with children is around 45%.

Cluster 4 contains the largest number of clients with 3375 and is similar to cluster 2 when it comes to customer value and portfolio management. However, they have a very low annual salary when compared to the average client and the majority of the individuals have a BSc/MSc. Most of them also have children.

Targeting both these groups may be done similarly to cluster 0 and 1 but targeting more on the motor insurance policies. These insurances average a lower claim rate thus creating more value for the company than the health insurance.

Cluster 3 has clients with above average value for our company and contain 1026 individuals. Their portfolio on average consists of very above average house, work and life insurance premium payments, quite below average motor insurance payments and average health insurance premiums. Their average salary is quite below average and their education mostly consists of high school. Most of the individuals in the cluster also have children.

Cluster 5 contains the most valuable clients of our company. They have a similar portfolio strategy as our customer in cluster 3 but they have a higher average salary than most our customers. Their education mostly consists of high school and most do not have children. These individuals also have the lowest claims rate as would be expected due to their main insurances (life and household) not being claimed as frequently when compared to health insurances.

To target the individuals in both these clusters a marketing strategy aimed at selling motor insurance may be developed as it seems to be the only category that they do not consider into their portfolio as much.

4.6. Cluster Visualization using t-SNE

We also decided to use t-SNE. This statistical method allows us to better visualize our data, understand certain subtleties in its distribution and provides us a clearer visualization of our clusters. As we can

see in figure F4.19, the data is not neatly separated (changing the parameter Perplexity did not alter significantly). However, it is possible to see that the clusters are positioned in different places in the space (although fuzzy). In addition, we can say specifically that there seems to be a certain overlap between the grey and purple clusters and the dark blue clusters is very spread out (which is reasonable considering it is one of the clusters with the most observations).

4.7. Assess Feature Importance and Classify Outliers

Afterwards, we determined the R^2 for each variable in our clustering solution. 'PremMotor' and 'AnnualSal' had the highest values while 'CustValYear' was the least significant. Then, we trained a Decision Tree Classifier using the variables we used for the clustering as independent variables and final cluster labels as the target variable. The objective is to see its accuracy and find the feature importance. We got an accuracy of 86.10% using *max_depth=4* (used it taking into consideration we want a simple and interpretable model and comparing the results we got in accuracy) and we can see the tree in figure F4.20. In terms of feature importance, both 'PremMotor' and 'AnnualSal' were the most important and 'PremHealth', 'PremLife', 'HighSchool', 'PhD' and 'CustValYear' were not used in defining the rules for the decision tree. We also trained a K-nearest Neighbors Classifier to see if we would have a higher accuracy, which we did - 87.89% - with *n_neighbors=6,* decided by comparing the results we got in accuracy and taking into consideration what would be a reasonable value. In the end, we joined the two outliers datasets (because we removed outliers two times in our project) and predicted the labels of the outliers using K-nearest Neighbors Classifier. We had to scale the data and create/remove features in order to join them and prepare the dataset for the classification. The majority went to the third cluster.

5. Conclusion

To begin our project, we analyzed the provided insurance client data to identify any wrong data types, missing values, inconsistencies, and outliers. We then applied various techniques to rectify the issues we found and the removed outliers were reintroduced after our clustering model was built.

During the pre-processing phase, we also created new variables, selected the most important and performed data normalization.

We divided our data set into two perspectives: Demographic information about the customer base and client purchase behaviors. Afterwards we applied several clustering methods: K-Prototypes for the demographic data and K-Means, Hierarchical Clustering and K-means, Gaussian Mixture Model, Self-Organizing Maps and K-means and Self-Organizing Maps and Hierarchical Clustering for the behavioral perspective. Finally, we merged the best solutions from both perspectives and ended up with 6 distinct groups of clients.

The final clusters showed a relatively balanced distribution on the bigger groups, although some clusters contained quite less individuals as would be expected when trying to differentiate the most valuable clients. The profiles were clear enough that it was possible to create marketing strategy suggestions to target each cluster. We were also able to identify the companies' most valuable clients in contrast with others, who must be monetized in a different way.

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7. Appendix

7.1 Tables

7.1.1 Data Exploration

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
FirstPolYear	10266.0	NaN	NaN	NaN	1991.062634	511.267913	1974.0	1980.0	1986.0	1992.0	53784.0
BirthYear	10279.0	NaN	NaN	NaN	1968.007783	19.709476	1028.0	1953.0	1968.0	1983.0	2001.0
EducDeg	10279	4	b'3 - BSc/MSc'	4799	NaN	NaN	NaN	NaN	NaN	NaN	NaN
MonthSal	10260.0	NaN	NaN	NaN	2506.667057	1157.449634	333.0	1706.0	2501.5	3290.25	55215.0
GeoLivArea	10295.0	NaN	NaN	NaN	2.709859	1.266291	1.0	1.0	3.0	4.0	4.0
Children	10275.0	NaN	NaN	NaN	0.706764	0.455268	0.0	0.0	1.0	1.0	1.0
CustMonVal	10296.0	NaN	NaN	NaN	177.892605	1945.811505	-165680.42	-9.44	186.87	399.7775	11875.89
ClaimsRate	10296.0	NaN	NaN	NaN	0.742772	2.916964	0.0	0.39	0.72	0.98	256.2
PremMotor	10262.0	NaN	NaN	NaN	300.470252	211.914997	-4.11	190.59	298.61	408.3	11604.42
PremHousehold	10296.0	NaN	NaN	NaN	210.431192	352.595984	-75.0	49.45	132.8	290.05	25048.8
PremHealth	10253.0	NaN	NaN	NaN	171.580833	296.405976	-2.11	111.8	162.81	219.82	28272.0
PremLife	10192.0	NaN	NaN	NaN	41.855782	47.480632	-7.0	9.89	25.56	57.79	398.3
PremWork	10210.0	NaN	NaN	NaN	41.277514	51.513572	-12.0	10.67	25.67	56.79	1988.7

T2.1: Description of the data

7.1.2 Demographic perspective

Cluster (with 2 in total)	AnnualSal	HighSchool	BSc/MSc	PhD	Children
0	0.954801	0.300634	0.562075	0.068763	0.447548
1	-0.710024	0.373575	0.422009	0.071403	0.910434

Cluster (with 3 in total)	AnnualSal	HighSchool	BSc/MSc	PhD	Children
0	-0.055344	0.151047	0.720339	0.060070	0.919990
1	-1.105940	0.557178	0.206465	0.064303	0.876955
2	1.166248	0.391566	0.428715	0.089692	0.269746

T4.1: Cluster centroids

7.1.3 Behavioral perspective

Cluster	CustValYear	PremMotor	PremHousehold	PremHealth	PremLife	PremWork
0	0.364421	-1.185315	1.277659	0.057003	1.244068	1.216964
1	0.004246	0.985648	-0.555794	-0.807881	-0.584453	-0.581570
2	-0.209138	-0.407131	-0.107694	0.841830	-0.018632	-0.023078

T4.2: Cluster centroids with K-Means

Final Cluster	CustValYear	PremMotor	PremHousehold	PremHealth	PremLife	PremWork
0	0.497209	-1.254656	1.740194	0.014431	1.110577	1.156776
1	-0.125750	-0.403983	-0.066865	0.639917	0.156674	0.113240
2	-0.035761	1.061580	-0.608487	-0.858989	-0.646918	-0.619210

T4.3: Cluster centroids with Hierarchical clustering + k-Means

Cluster	CustValYear	PremMotor	PremHousehold	PremHealth	PremLife	PremWork
0	-0.124177	-0.220948	-0.075742	0.493507	-0.009098	-0.025486
1	-0.036737	0.991024	-0.657078	-0.643149	-0.692554	-0.674137
2	0.255627	-1.169827	1.146204	0.170594	1.155211	1.128865

T4.4: Cluster centroids with Gaussian Mixture Model

Final Cluster	CustValYear	PremMotor	PremHousehold	PremHealth	PremLife	PremWork
0	-0.188902	-0.488569	0.028207	0.737065	0.161299	0.113829
1	0.521182	-1.316222	1.489007	0.046436	1.274230	1.406354
2	-0.005213	0.949765	-0.568222	-0.721444	-0.592973	-0.600612

T4.5: Cluster centroids with Self-organizing Map and K-mean

Final Cluster	CustValYear	PremMotor	PremHousehold	PremHealth	PremLife	PremWork
0	-0.021435	0.719938	-0.463904	-0.468096	-0.485453	-0.516891
1	0.529955	-1.232049	1.437899	0.051273	1.342144	1.096773
2	-0.283774	-0.696040	0.045818	0.939100	0.200443	0.374859

T4.6: Cluster centroids with Self-organizing Map and Hierarchical Clustering

7.1.4 Merged perspectives

Final Cluster	PremMotor	PremHousehold	PremHealth	PremLife	PremWork	CustValYear	AnnualSal	CustMonVal	ClaimsRate	YearsCustomer
0	-0.478217	0.037323	0.691224	0.177243	0.134113	-0.178630	-0.638591	-0.169621	0.176397	0.046630
1	-0.503381	0.015165	0.802651	0.138487	0.084808	-0.203599	1.197687	-0.205819	0.213735	0.002697
2	0.771704	-0.510699	-0.506679	-0.515631	-0.540014	0.006479	1.121003	-0.006360	-0.125534	-0.028051
3	-1.380100	1.528988	-0.005471	1.406497	1.530465	0.441911	-1.175310	0.459956	-0.085684	-0.036039
4	0.998883	-0.584089	-0.780688	-0.614308	-0.617328	-0.008438	-0.180489	-0.008962	-0.147730	0.011735
5	-1.110125	1.360011	0.213910	0.847484	1.005923	0.776943	1.126781	0.718830	-0.331478	-0.205556

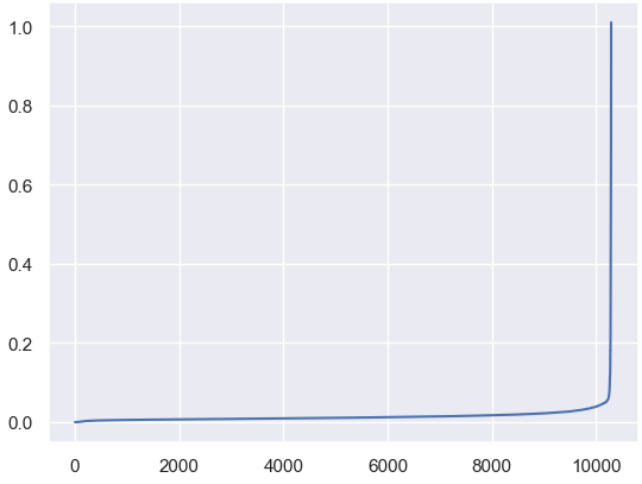
T4.7: Cluster centroids with merged perspectives (metric features only)

Final Cluster	HighSchool	BSc/MSc	PhD	Children
0	0.392283	0.468650	0.037379	0.888666
1	0.409431	0.446233	0.051754	0.182289
2	0.291085	0.490870	0.187970	0.454350
3	0.452242	0.170565	0.003899	0.791423
4	0.227852	0.634963	0.097481	0.945481
5	0.588050	0.150943	0.009434	0.207547

T4.8: Cluster centroids with merged perspectives (non-metric features only)

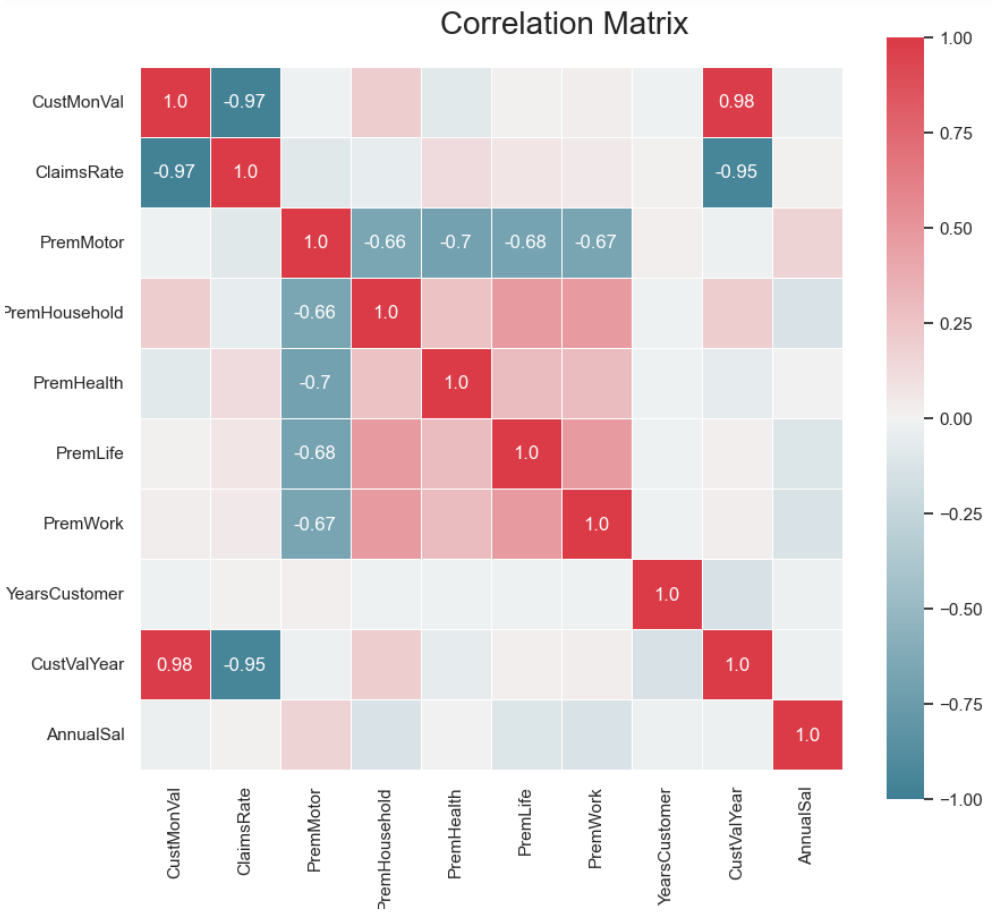
7.2 Figures

7.2.1 DBSCAN



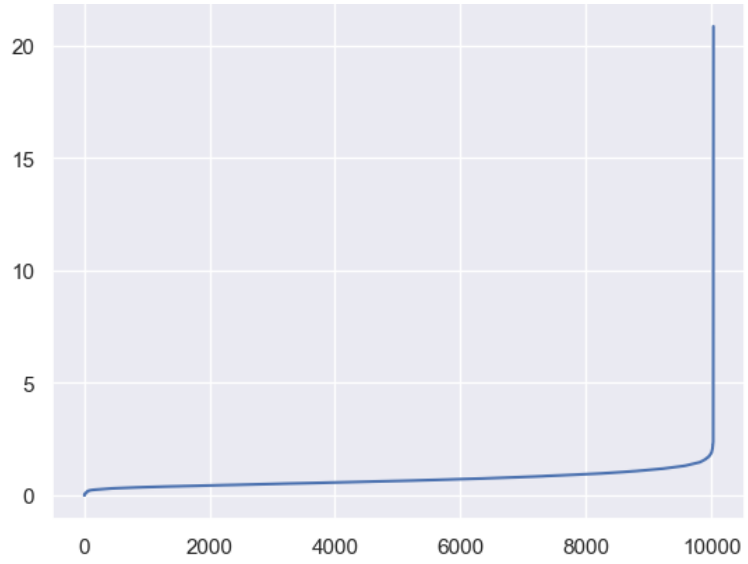
F3.1: K-distance elbow plot

7.2.2 Association Measures

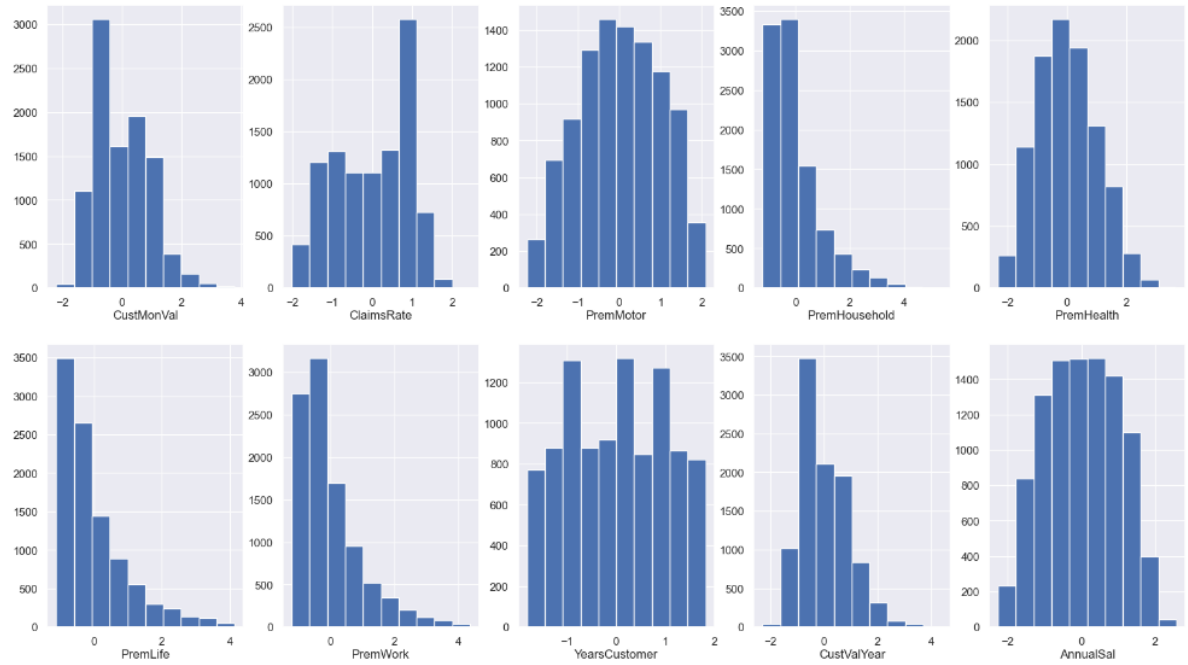


F3.2: Initial correlation matrix

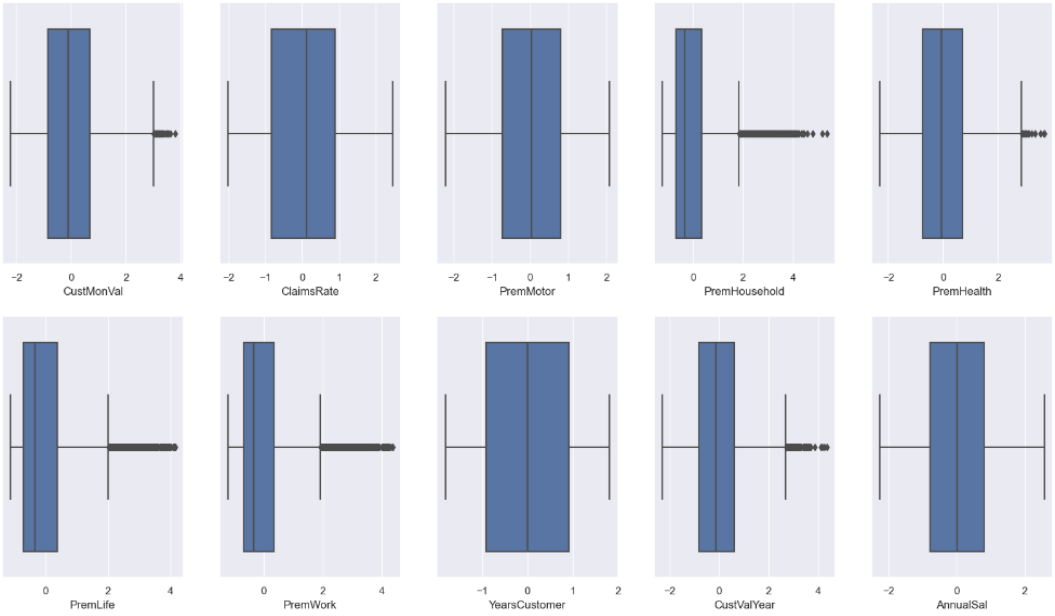
7.2.3 Detection of the Remaining Outliers in our Dataset (using DBSCAN)



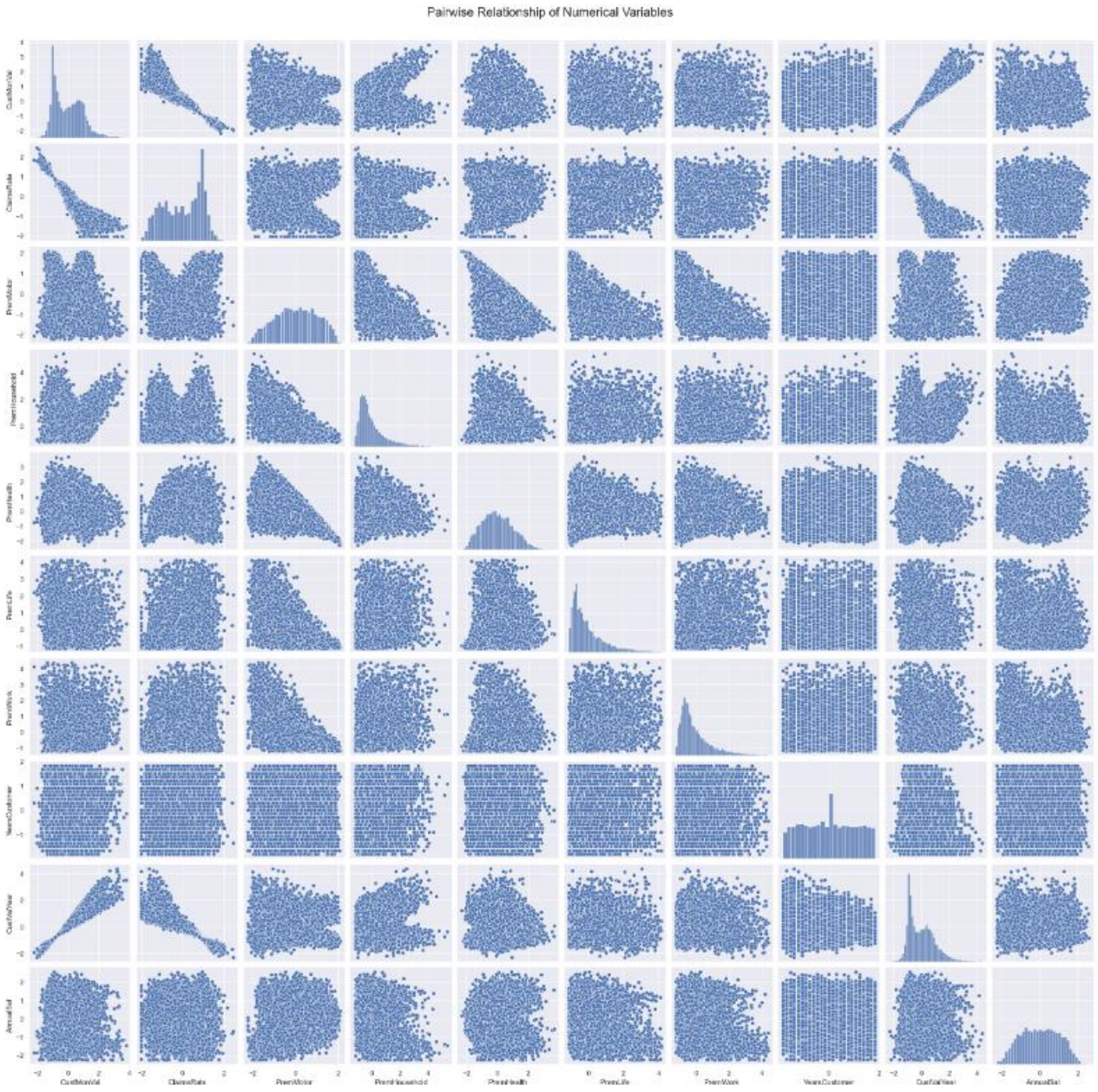
F3.3.1: K-Distance elbow plot



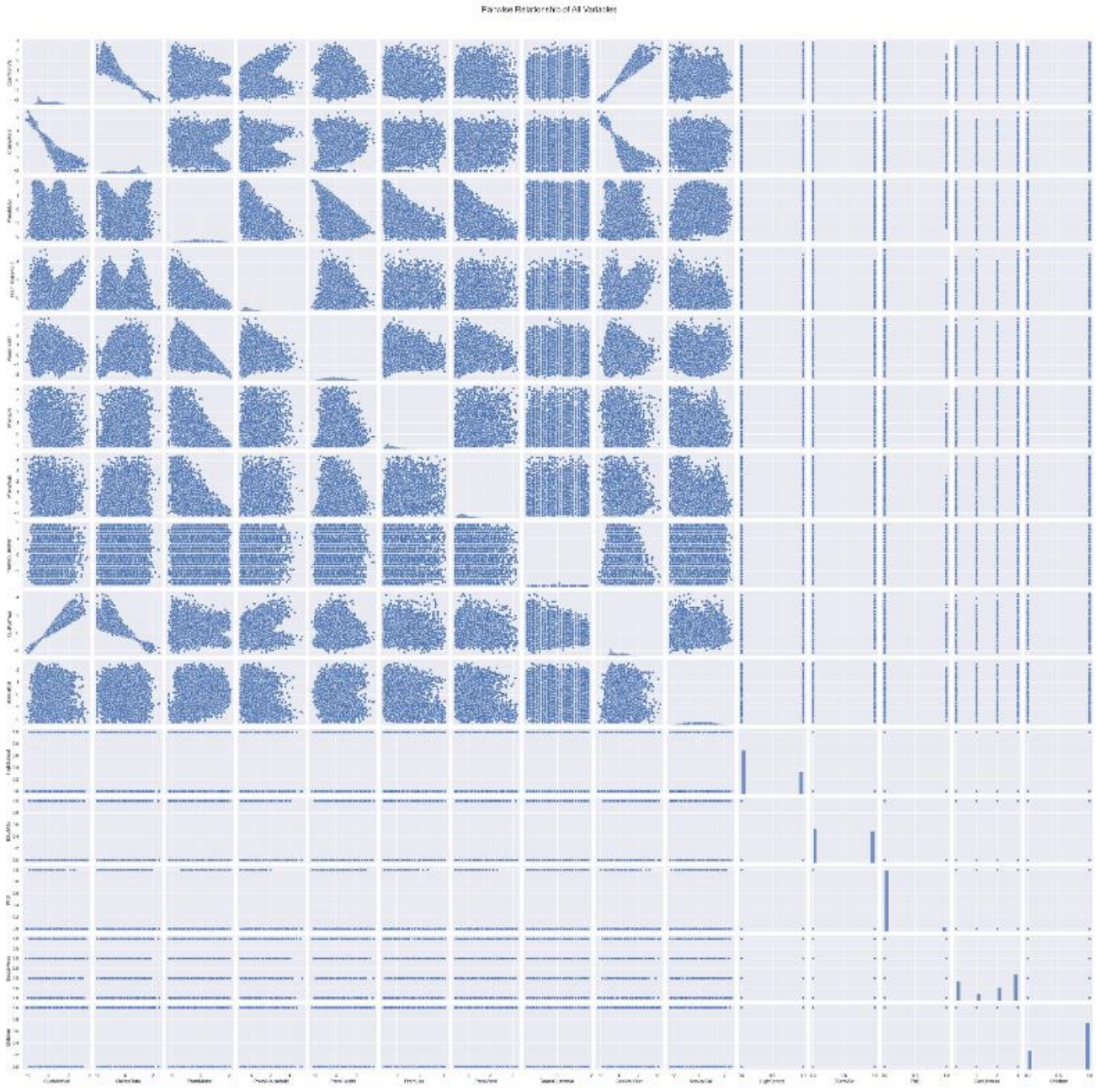
F3.3.2: Histograms of the preprocessed data



F3.3.3: Box Plots of the preprocessed data

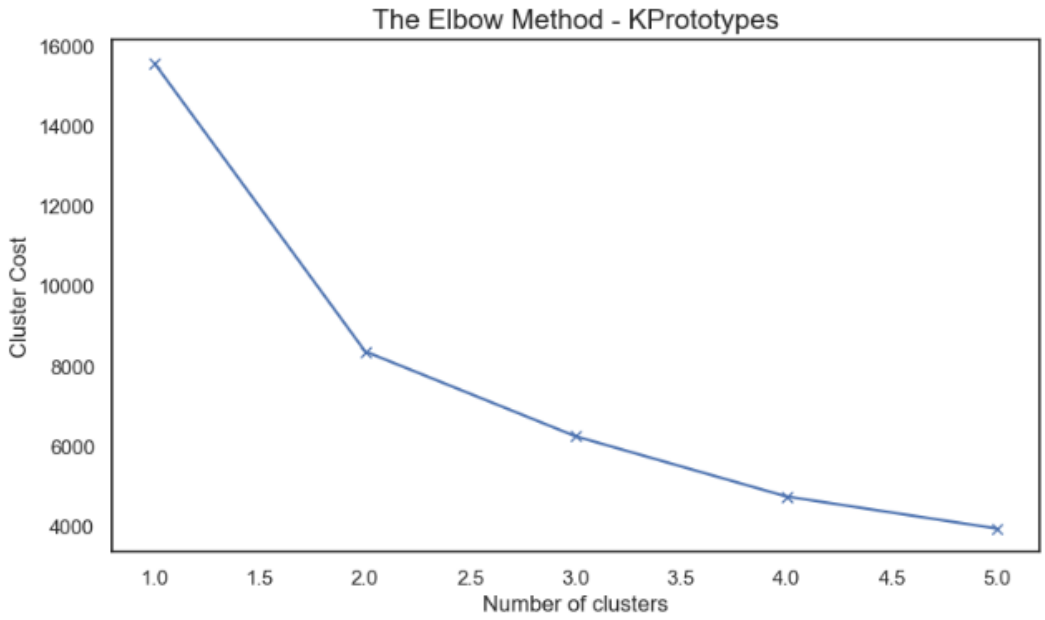


F3.3.4: Pairwise relationships of numerical data

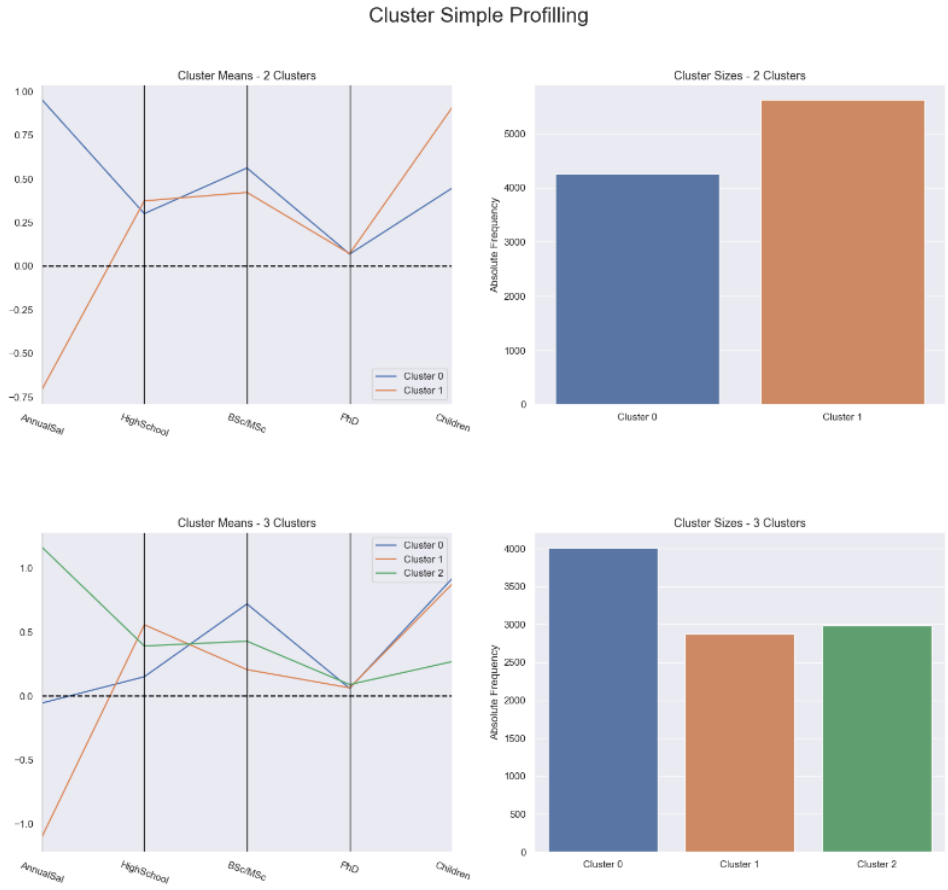


F3.3.5: Pairwise relationships of all the data

7.2.4 Demoraphic perspective: K-prototypes

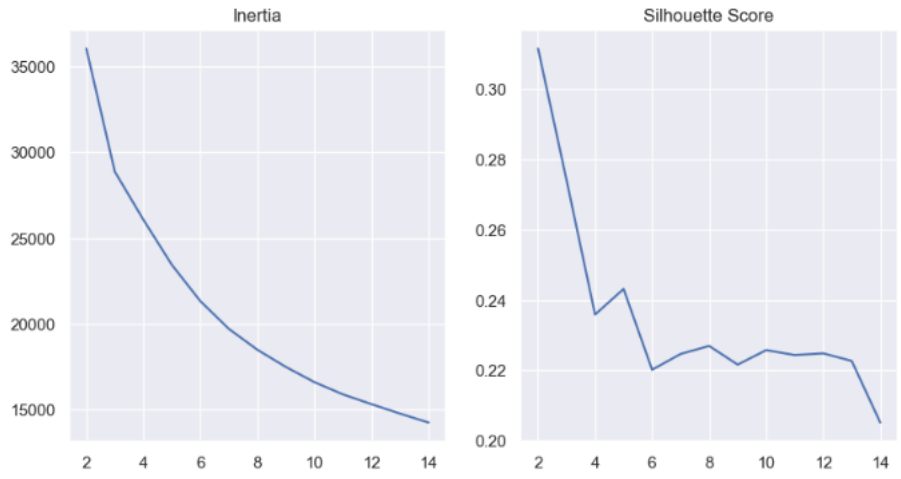


F4.1: K-prototypes elbow plot



F4.2: K-prototypes cluster simple profiling

7.2.5 Behavorial perspective: K-means



F4.3: Inertia and Silhouette score plots with K-means

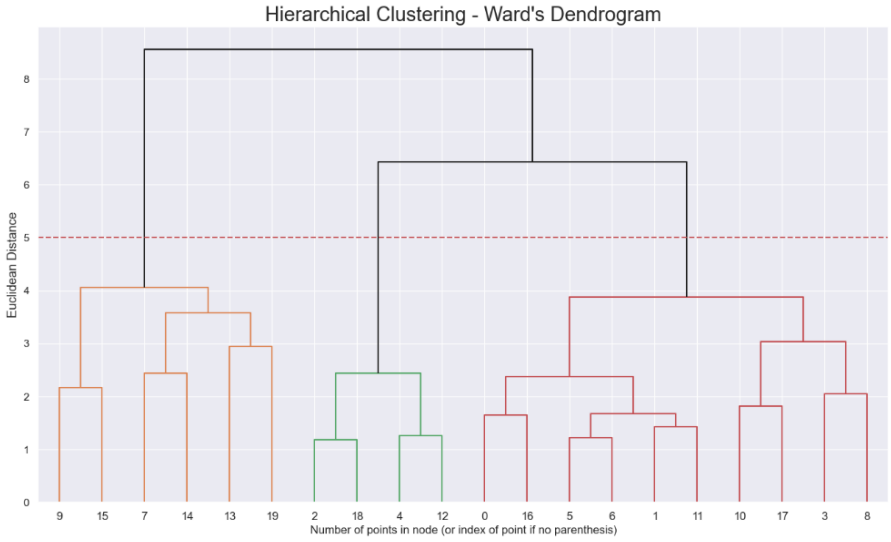


F4.4: Simple cluster profiling with K-means

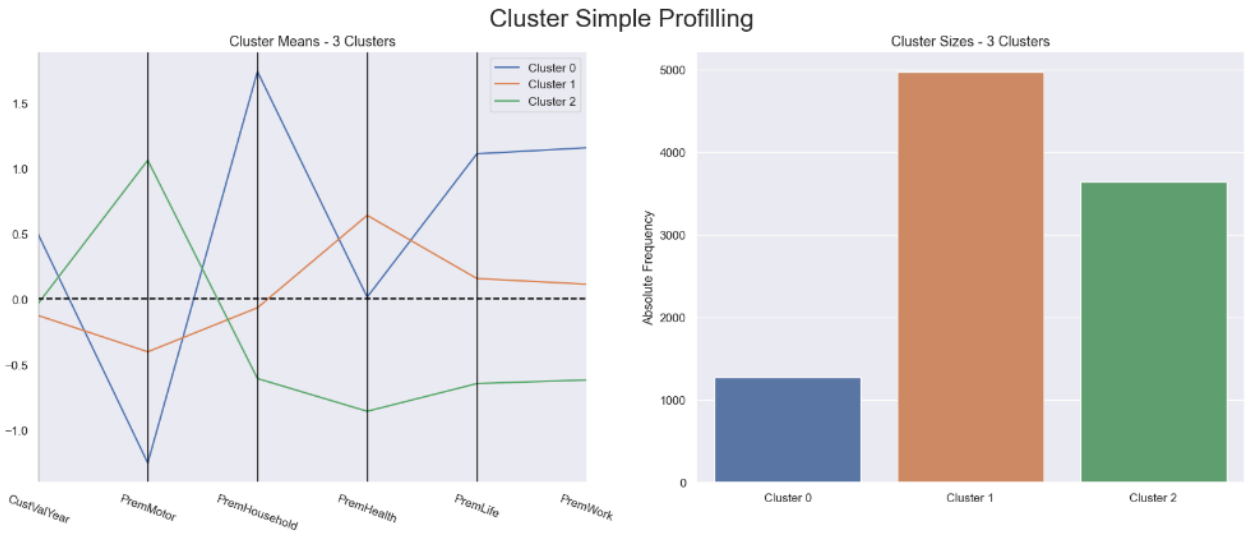
7.2.6 Behavorial perspective: K-means and Hierarchical Clustering



F4.5: R^2 values with different numbers of clusters with K-means and Hierarchical Clustering

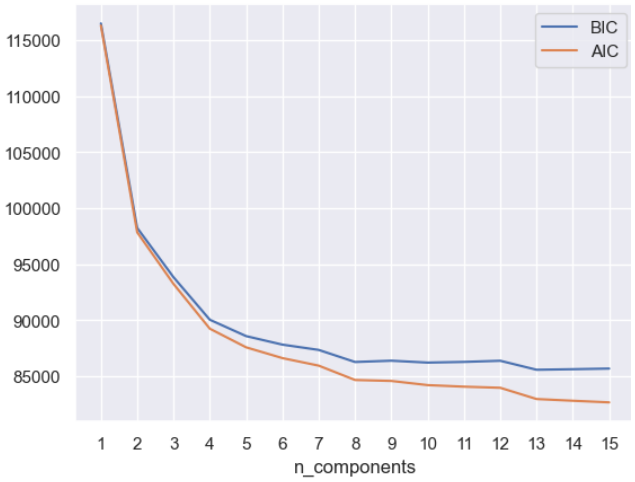


F4.6: Ward's Dendrogram with K-means and Hierarchical Clustering

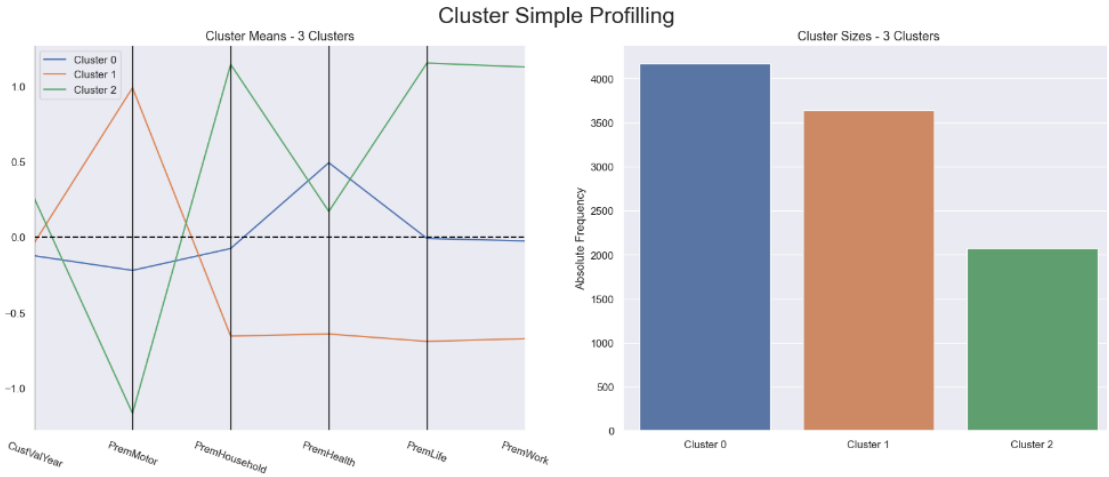


F4.7: Simple cluster profiling with K-means and Hierarchical Clustering

7.2.7 Behavioral perspective: Gaussian Mixture Model

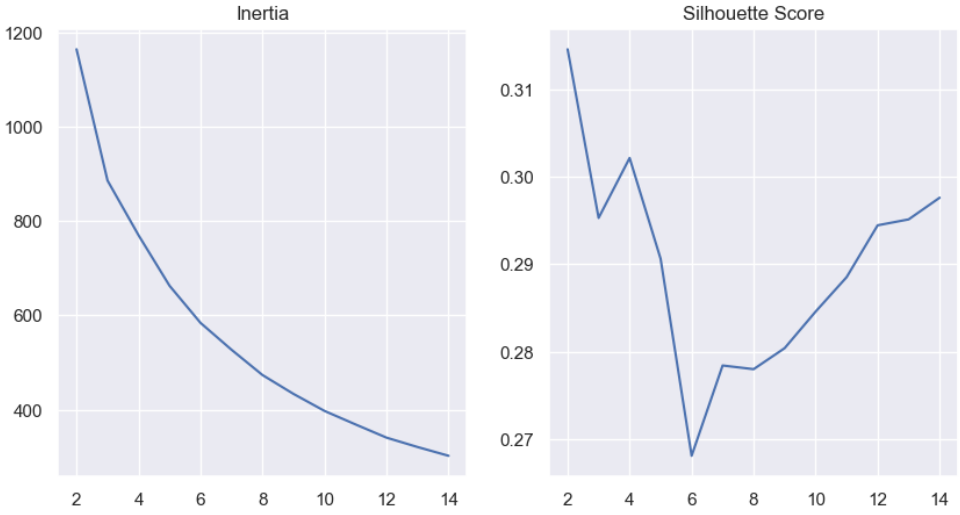


F4.8: Plot for selecting the number of components based on AIC and BIC with Gaussian Mixture Model

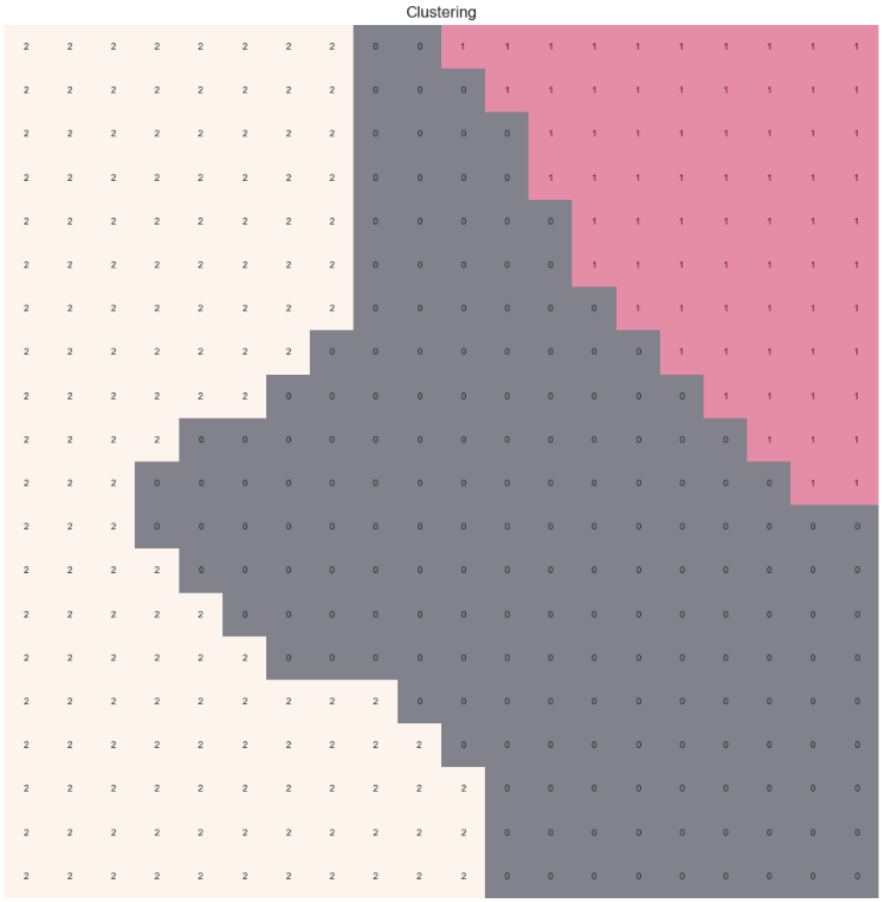


F4.9: Simple cluster profiling with Gaussian Mixture

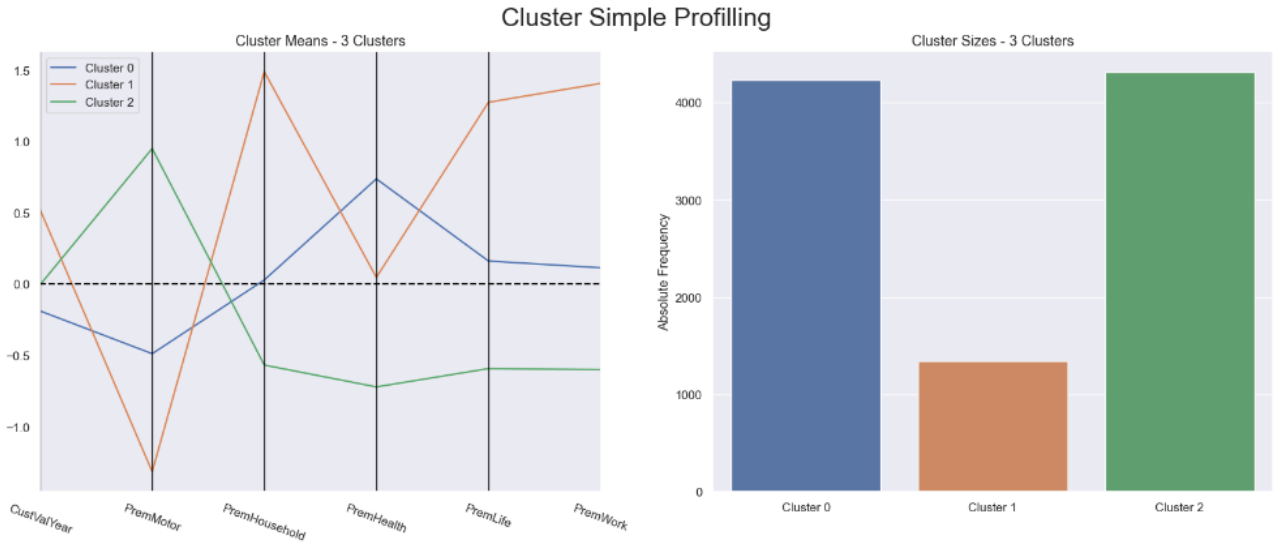
7.2.8 Behavioral perspective: Self-organizing Map and K-means



F4.10: Inertia and Silhouette score plots with Self-organizing Map and K-means

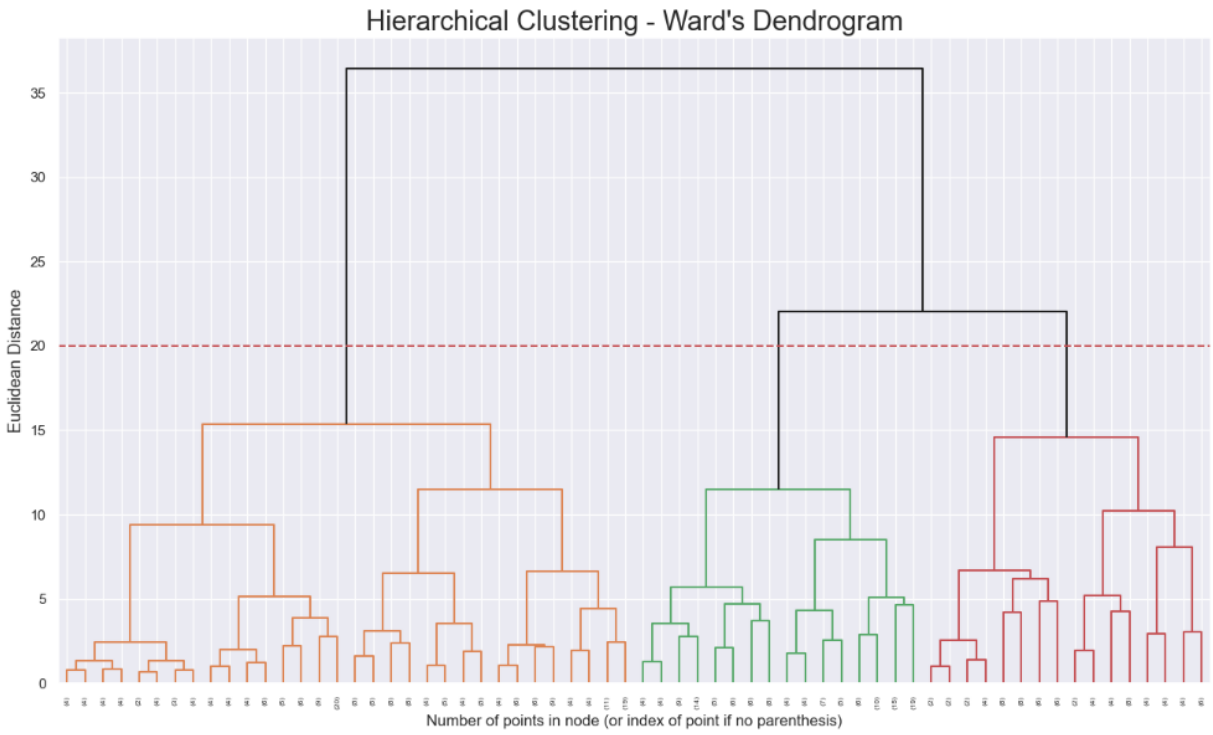


F4.11: Hit Map for clustering visualization with Self-organizing Map and K-means

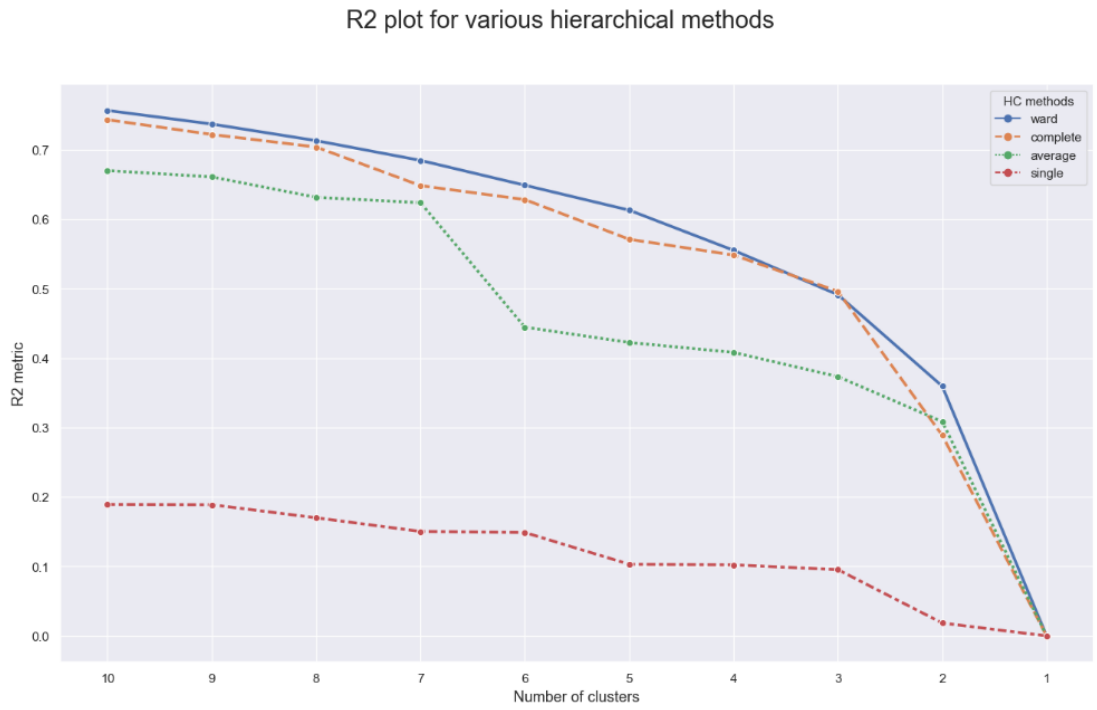


F4.12: Simple cluster profiling with Self-organizing Map and K-means

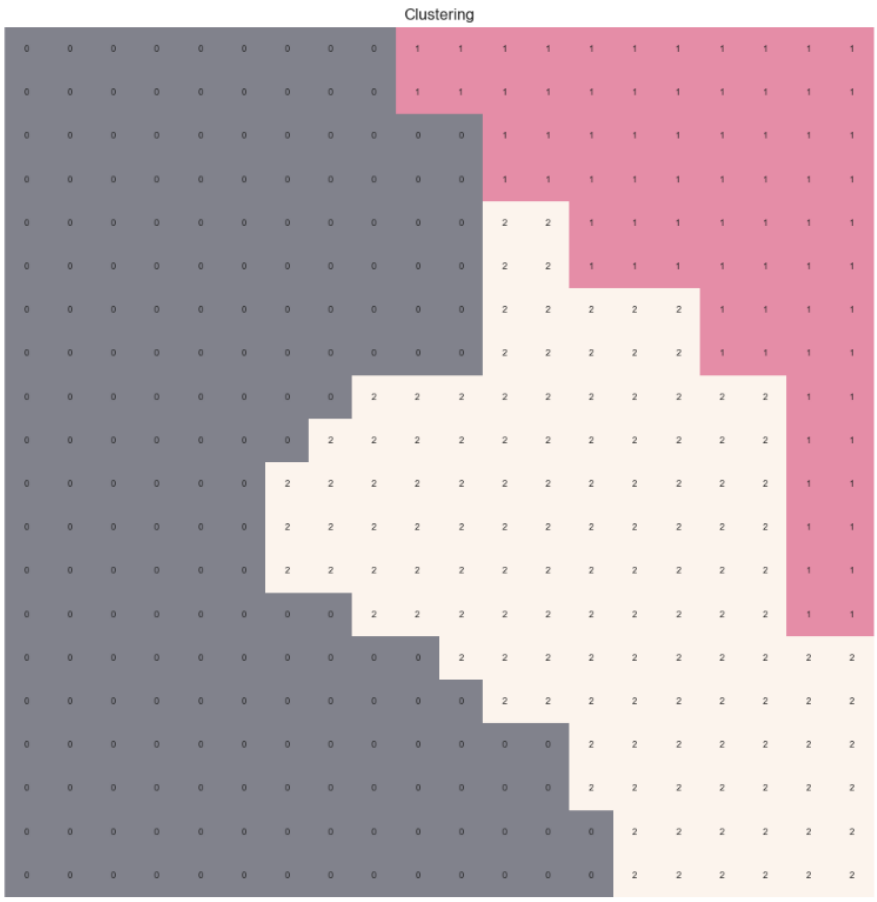
7.2.9 Behavioral perspective: Self-organizing Map and Hierarchical Clustering



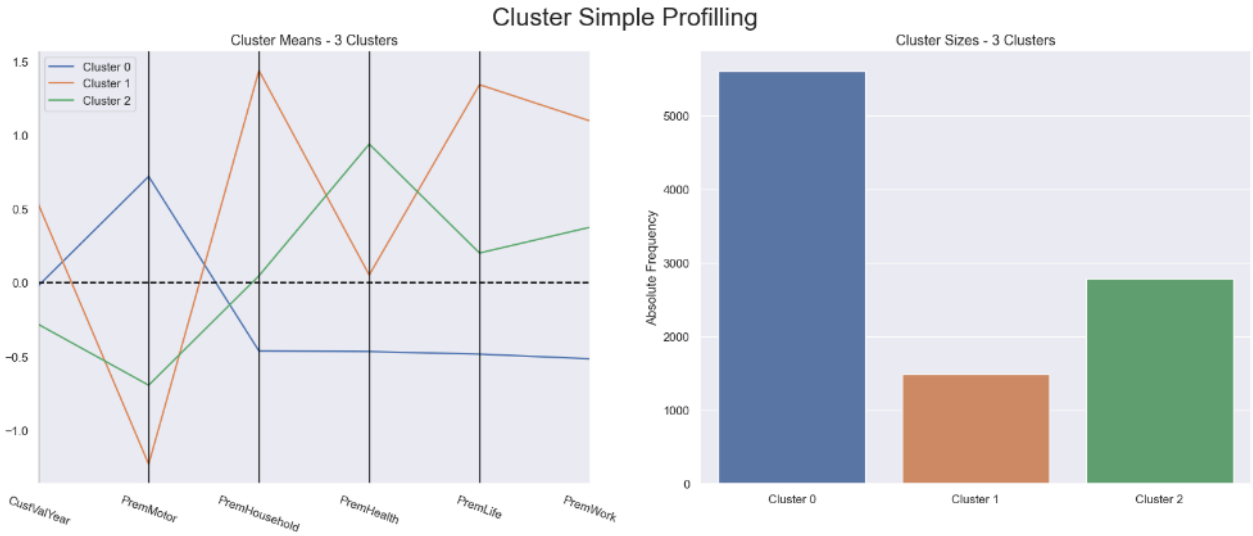
F4.13: Ward's Dendrogram with Self-organizing Map and Hierarchical Clustering



F4.14: R^2 values with different numbers of clusters with Self-organizing Map and Hierarchical Clustering

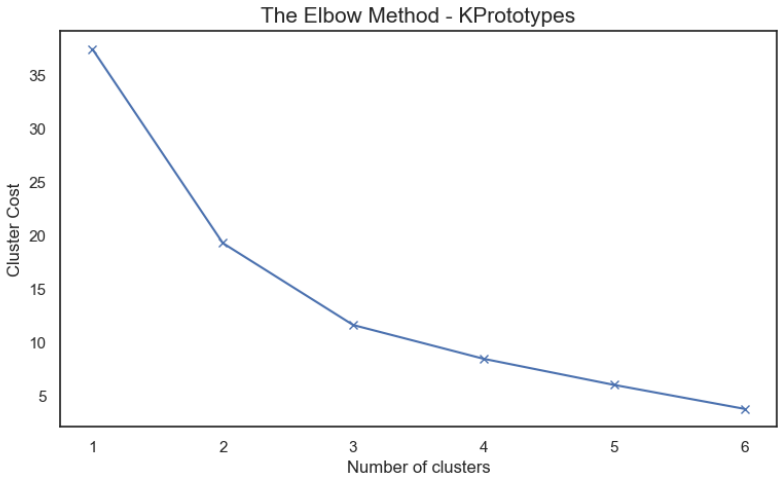


F4.15: Hit Map for clustering visualization with Self-organizing Map and Hierarchical Clustering



F4.16: Simple cluster profiling with Self-organizing Map and Hierarchical Clustering

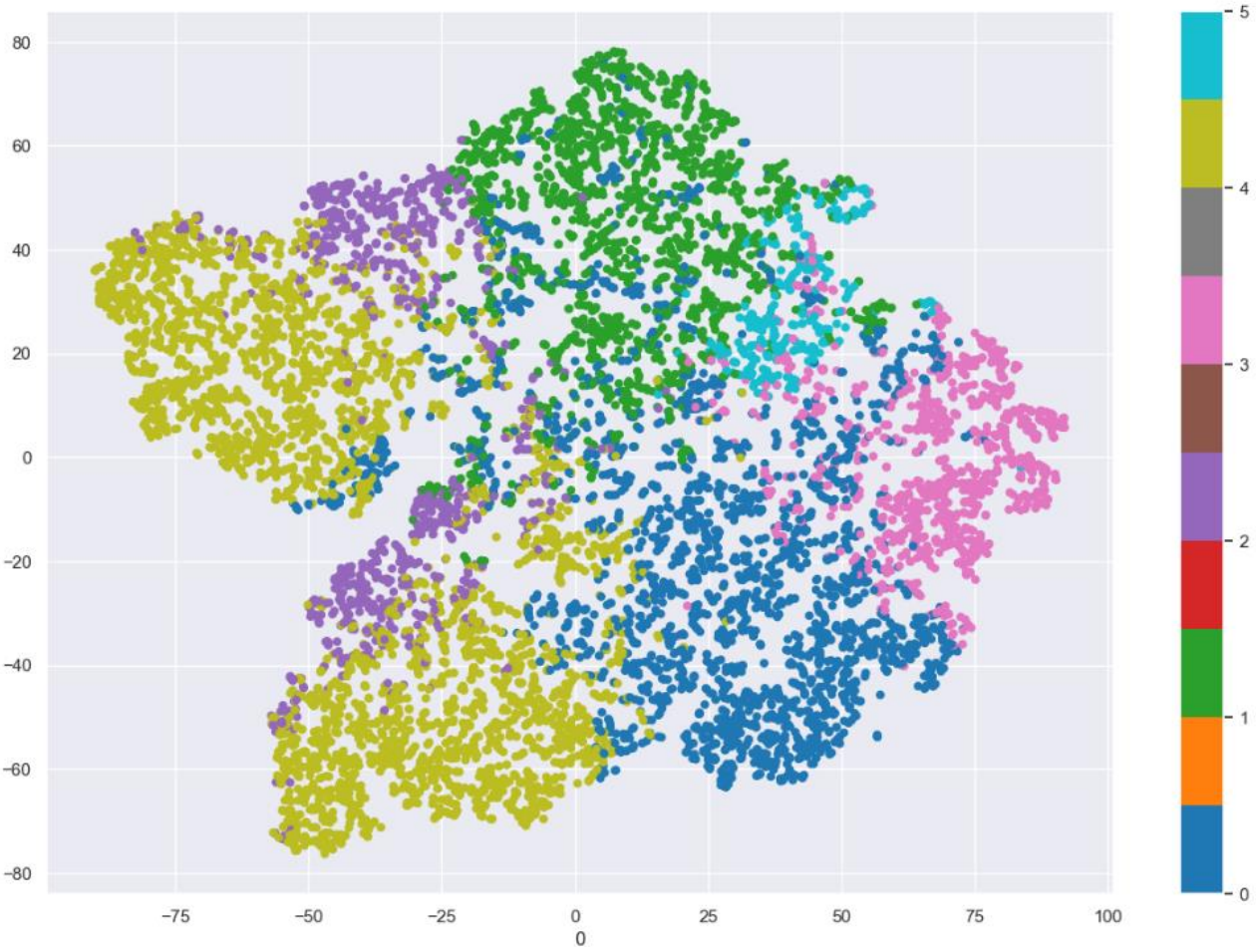
7.2.10 Merged Perspectives



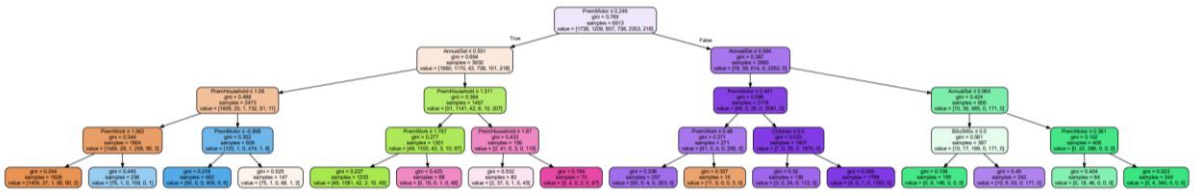
F4.17: Calculation of cluster cost for each number of clusters with K-prototypes



F4.18: Simple cluster profiling for the final solution



F4.19: Final cluster visualization using t-SNE



F4.20: Decision Tree